Karan Parmar

Software Development Engineer

Pune kp5336464@gmail.com 9356258692 leetcode KaranParmar19 Karan19

Education

Lovely Professional University *2023 – 2027*
B-Tech in Computer Science and Engineering
- CGPA: **8.03/10**
- Courses: **DSA, DBMS, Operating System, Computer Networking**

Technical Skills

Languages: C, C++, JavaScript, Python
Backend: Node.js, Express.js, RESTful APIs, JWT, Bcrypt
Frontend: React, Tailwind CSS, Framer
Databases: MongoDB, MySQL, DBMS
Deployment: Google Cloud, Render, Vercel
Tools & DevOps: Git, GitHub, Postman,Dotenv, Nodemonl
Soft Skills: Ownership, Critical Thinking, Strategic Planning

Projects

Voxa - Real Time Collaborative Workspace (MERN + WebRTC) *Github-link* *Live*
- Engineered a full-stack, real-time collaboration platform featuring an infinite whiteboard (Fabric.js), a live code editor (Monaco), and integrated P2P voice communication.
- Implemented **sub-50ms** state synchronization across clients using WebSockets (Socket.IO) with custom render throttling to prevent UI lag on slow network connections.
- Optimized frontend performance by decoupling heavy canvas state from standard React component state, eliminating unnecessary re-renders during high-frequency drawing events.
- Built a secure REST API and WebRTC signaling server with Express and Node.js, featuring resilient Google OAuth and JWT-based authentication for persistent workspaces in MongoDB.

MemoryViz - Virtual Memory Simulation (ReactJs + D3.js + Python) *Github-link*
- Accepts user-defined parameters to dynamically execute and visualize the Virtual-to-Physical address Translation process,
- Bridges the gap between abstract OS concepts and practical understanding, serving as a powerful demonstration of proficiency in **system-level programming**, data visualization, and educational tool development.
- Features modules that graphically demonstrate and compare key memory management techniques: Paging, Segmentation and Combined Scheme.

Achievement/Certificates

- Solved **300+ problems** on **LeetCode** demonstrating rigorous mastery of algorithmic complexity and **competitive programming** techniques (Dynamic Programming, Graph Theory and many more).
- Completed the **Data Structure** and **Algorithm** by SkillStone a Grazitti interative initiative, issued in Aug 2025.